LM_HW3

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2022-10-28

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1 Problem 1

The significance level *alpha* is set as 0.05.

```
dat1 = read.table("uswagesall.txt", header = TRUE)
```

1.1 a

(i)

Full model :

$$\text{wage} = \beta_0 + \beta_1 \,\text{educ} + \beta_2 \,\text{exper} + \epsilon \tag{1}$$

Reduced model :

$$\text{wage} = \beta_0 + \epsilon \tag{2}$$

Null and alternative hypotheses:

$$\begin{cases} H_0 : \beta_1 = \beta_2 = 0 \\ H_a : \beta_1 \text{ or } \beta_2 = 0 \text{ is not equal to zero} \end{cases} \tag{3}$$

```
fit = lm(wage~educ+exper, data = dat1)
summary(fit)
```

```
##
## Call:
## lm(formula = wage ~ educ + exper, data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1136.1  -220.8   -48.3   154.5 18156.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -385.0834    13.2428  -29.08   <2e-16 ***
## educ          60.8964     0.8828   68.98   <2e-16 ***
## exper         10.6057     0.1957   54.19   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 411.5 on 28152 degrees of freedom
## Multiple R-squared:  0.1768, Adjusted R-squared:  0.1768
## F-statistic:  3024 on 2 and 28152 DF,  p-value: < 2.2e-16
```

Report:

According to the R output, the test statistic is 3024 and the p-value is very closed to zero. Hence, we reject H_0 and conclude that there is strong evidence saying that either years of education or experience has significant predictive value for wages.

(ii)

Full model :

$$\text{wage} = \beta_0 + \beta_1 \,\text{educ} + \beta_2 \,\text{exper} + \epsilon \tag{4}$$

Reduced model :

$$\text{wage} = \beta_0 + \beta_2 \,\text{exper} + \epsilon \tag{5}$$

Null and alternative hypotheses:

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_a : \beta_1 \neq 0 \end{cases} \tag{6}$$

```
fit = lm(wage~educ+exper, data = dat1)
summary(fit)
```

```
##
## Call:
## lm(formula = wage ~ educ + exper, data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1136.1  -220.8   -48.3   154.5 18156.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -385.0834    13.2428  -29.08   <2e-16 ***
## educ          60.8964     0.8828   68.98   <2e-16 ***
## exper         10.6057     0.1957   54.19   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 411.5 on 28152 degrees of freedom
## Multiple R-squared:  0.1768, Adjusted R-squared:  0.1768
## F-statistic:  3024 on 2 and 28152 DF,  p-value: < 2.2e-16
```

Report:

According to the R output, the test statistic is 68.98 and the p-value is very closed to zero. Hence, we reject H_0 and conclude that there is strong evidence saying that years of education has significant predictive value for wages when experience is also included in the model.

(iii)

Full model :

$$\text{wage} = \beta_0 + \beta_1 \,\text{educ} + \epsilon \tag{7}$$

Reduced model :

$$\text{wage} = \beta_0 + \epsilon \tag{8}$$

Null and alternative hypotheses:

$$\begin{cases} H_0 : \beta_1 = 0 \\ H_a : \beta_1 \neq 0 \end{cases} \tag{9}$$

```r
fit = lm(wage~educ, data = dat1)
summary(fit)
```

```
##
## Call:
## lm(formula = wage ~ educ, data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
##  -786.0  -264.9   -54.8   174.5 18035.1
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -12.8281    11.8970  -1.078    0.281
## educ         47.1810     0.8888  53.085   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 432.4 on 28153 degrees of freedom
## Multiple R-squared:  0.09099,    Adjusted R-squared:  0.09096
## F-statistic:  2818 on 1 and 28153 DF,  p-value: < 2.2e-16
```

Report:

According to the R output, the test statistic is 53.085 and the p-value is very closed to zero. Hence, we reject H_0 and conclude that there is strong evidence saying that years of education has significant predictive value for wages when experience is not included in the model.

1.2 b

One additional year of experience would on average increase 10.6057 dollars of weekly wages for people who have the same years of education.

1.3 c

(i)

No, we can not since we need the responses to be in the same scale for the two models so that the F-test can be applied. More specifically, the different scales let the two models belong to distinct model spaces which are disjoint and hence beyond the use of F-test.

(ii)

$$\text{wage} = \beta_0 + \beta_1 \,\text{educ} + \beta_2 \,\text{exper} + \epsilon \tag{10}$$

```r
fit1 = lm(wage~educ+exper, data = dat1)
summary(fit1)
```

```
##
## Call:
## lm(formula = wage ~ educ + exper, data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1136.1  -220.8   -48.3   154.5 18156.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -385.0834    13.2428  -29.08   <2e-16 ***
## educ          60.8964     0.8828   68.98   <2e-16 ***
## exper         10.6057     0.1957   54.19   <2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 411.5 on 28152 degrees of freedom
## Multiple R-squared:  0.1768, Adjusted R-squared:  0.1768
## F-statistic:  3024 on 2 and 28152 DF,  p-value: < 2.2e-16
```

$$\log(\text{wage}) = \beta_0 + \beta_1 \,\text{educ} + \beta_2 \,\text{exper} + \epsilon \tag{11}$$

```r
fit2 = lm(log(wage)~educ+exper, data = dat1)
summary(fit2)
```

```
##
## Call:
## lm(formula = log(wage) ~ educ + exper, data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2412 -0.3308  0.0888  0.4211  3.7032
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.4887875  0.0204402  219.60   <2e-16 ***
## educ        0.1013404  0.0013627   74.37   <2e-16 ***
## exper       0.0196442  0.0003021   65.02   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6352 on 28152 degrees of freedom
## Multiple R-squared:  0.2128, Adjusted R-squared:  0.2128
## F-statistic:  3806 on 2 and 28152 DF,  p-value: < 2.2e-16
```

Yes, we could say that Model (11) is a better fitting model than Model (10) in terms of R^2, since the R^2 is 0.1768 for Model (10) and 0.2128 for Model (11). Such fact implies that (11) explains higher proportion of variation in the response than Model (11) does.

1.4 d

```r
exp(coef(fit2))-1
```

```
## (Intercept)        educ       exper
## 88.01345449  0.10665326  0.01983842
```

One additional year of experience would on average increase 1.9 percentage of dollars of weekly wages for people who have the same years of education.

1.5 e

Full model :

$$\log(\text{wage}) = \beta_0 + \beta_1 \,\text{educ} + \beta_2 \,\text{exper} + \epsilon \tag{12}$$

Reduced model :

$$\log(\text{wage}) = \beta_0 + 0.1 \,\text{educ} + \beta_2 \,\text{exper} + \epsilon \tag{13}$$

Null and alternative hypotheses:

$$\begin{cases} H_0 : \beta_1 = 0.1 \\ H_a : \beta_1 \neq 0.1 \end{cases} \tag{14}$$

Test statistic in F-test

$$F = \frac{(RSS_{reduced} - RSS_{full})/(df_{full} - df_{reduced})}{RSS_{full}/df_{full}} \overset{H_0}{\sim} F_{1,28152}, \tag{15}$$

where df denotes the degrees of freedom for the residuals.

```
# Use general F-test
fit2 = lm(log(wage)~educ+exper, data = dat1)
full.rss = anova(fit2)[3, 2]

fit3 = lm((log(wage)-0.1*educ)~exper, data = dat1)
reduced.rss = anova(fit3)[2, 2]

F.statistic = (reduced.rss-full.rss)/(full.rss/28152)
p.value = 1-pf(F.statistic, 1, 28152)

c(F.statistic = F.statistic, p.value = p.value)
```

```
## F.statistic     p.value
##   0.9675405   0.3253028
```

Test statistic in t-test

$$T = \frac{\hat{\beta}_1 - 0.1}{\hat{SE}(\hat{\beta}_1)} \overset{H_0}{\sim} t_{28152}, \tag{16}$$

where df denotes the degrees of freedom for the residuals.

```
# use t-test
beta1 = coef(fit2)[2]
se.beta1 = summary(fit2)$coefficient[2,2]
t.statistic = as.numeric((beta1-0.1)/se.beta1)
p.value = 2*(1-pt(abs(t.statistic), 28152))

c(t.statistic = t.statistic, p.value = p.value)
```

```
## t.statistic     p.value
##   0.9836363   0.3253028
```

Report:

Since the p-values are greater than 0.05, we do not reject H_0 and conclude that there is no enough precision of the data for us to say that β_1 is significantly not equal to 0.1.

1.6 f

(i)

Model:

$$\log(\text{wage}) = \beta_0 + \beta_1 \,\text{educ} + \beta_2 \,\text{exper} + \epsilon \tag{17}$$

```r
newdata = dat1[1000*(1:28), ]
new.fit = lm(log(wage)~educ+exper, data = newdata)
summary(new.fit)
```

```
##
## Call:
## lm(formula = log(wage) ~ educ + exper, data = newdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.43154 -0.27358  0.05187  0.40237  0.91710
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.736873   0.510565   9.278 1.42e-09 ***
## educ        0.113308   0.035595   3.183  0.00387 **
## exper       0.004255   0.008418   0.505  0.61765
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6134 on 25 degrees of freedom
## Multiple R-squared:  0.2896, Adjusted R-squared:  0.2328
## F-statistic: 5.096 on 2 and 25 DF,  p-value: 0.01392
```

```r
c(reduced.R.squared = summary(new.fit)$r.squared, full.R.squared = summary(fit2)$r.squared)
```

```
## reduced.R.squared    full.R.squared
##          0.289628          0.212831
```

Report:

1. The reduced data version has the higher R^2.

2. A reduced data may not always have a higher R^2 than the full data. It is because that the meaning of R^2 is the proportion of variation in the response that can be explained by the predictors, which has nothing to do with the sample size. Instead, R^2 is usually much more related to the distribution of predictors.

(ii)

1. Years of education is statistically significant in the reduced data version.
2. This result in the reduced data version is different from that in the full data version where both the years of education and experience are statistically significant. Two possible reason is given as follows.

```
ggpairs(dat1, columns = 1:3)+theme_bw()
```

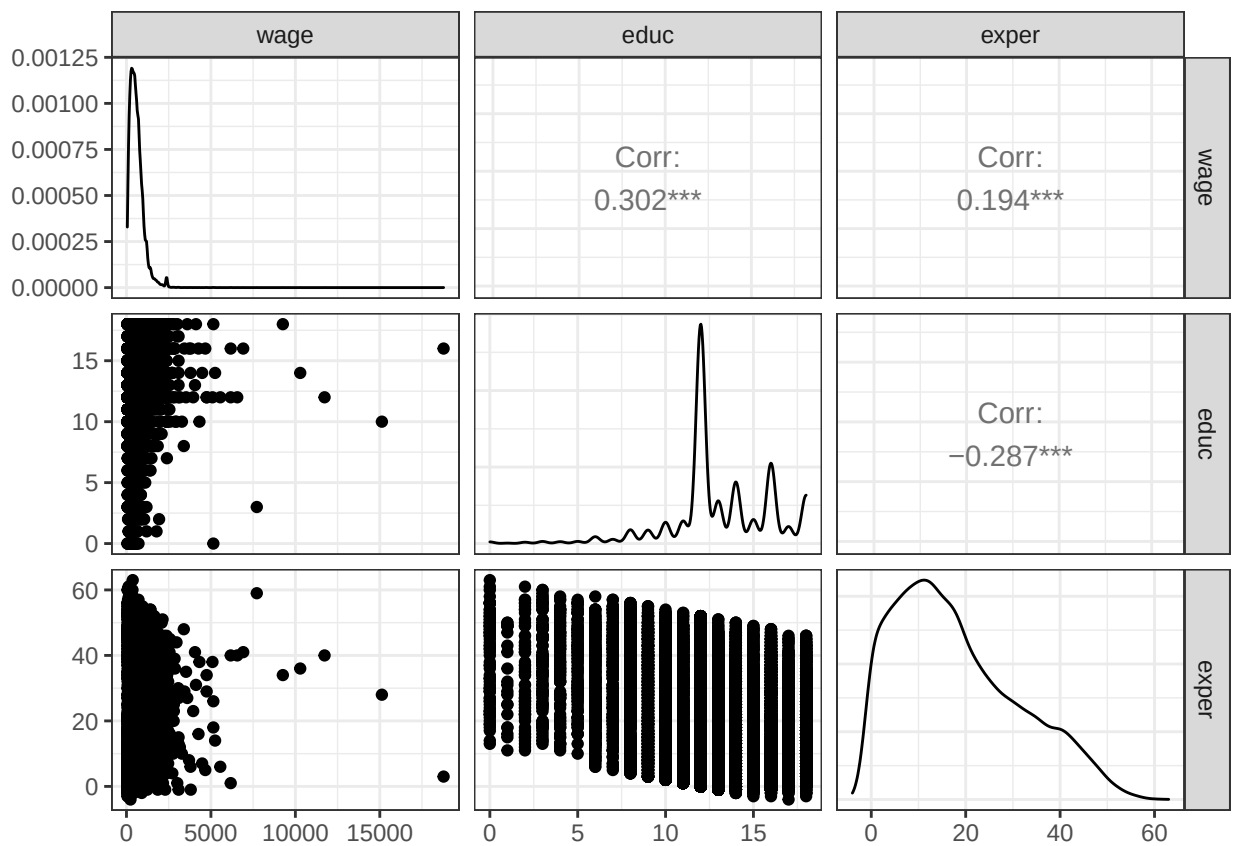

Figure 1: Relationships among the variables in full data

```
ggpairs(newdata, columns = 1:3)+theme_bw()
```

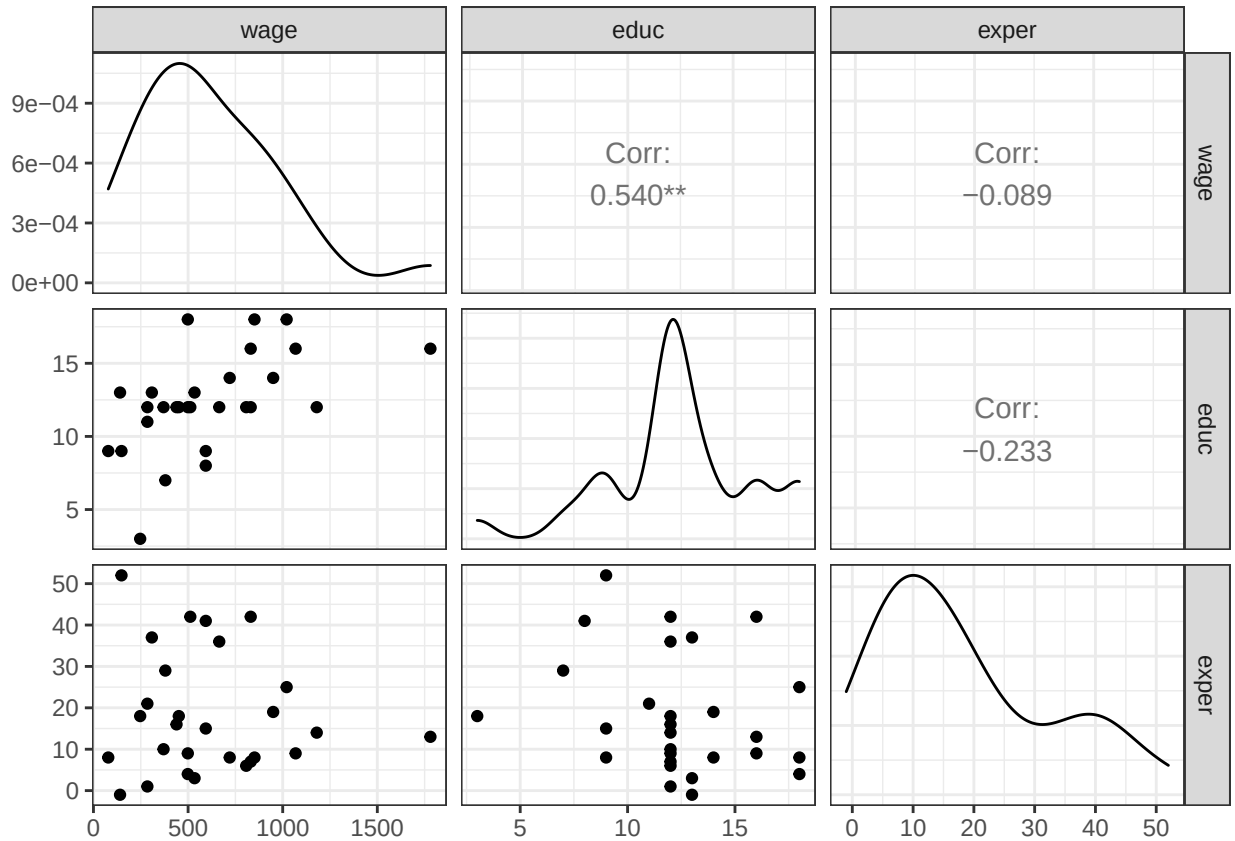

Figure 2: Relationships among the variables in reduced data

Possible reason 1: the two versions of data in fact carries different information

Figure 1 and 2 show the relationships among `wage`, `educ` and `exper` for the full data and the reduced data respectively. As one can see, the covariance structure of the three variables does not seem to be identical in the two versions of data according to the scatter plots and the correlation coefficients. In particular, all the three variables are significantly correlated with one another in the full data, yet only `educ` is significantly correlated with `wage` in the reduced data. Such fact implies that the reduced data may not carry the same information as the full data does. Hence, no wonder that there seems to be huge difference between the results for the two versions in terms of the significance of `exper`.

Possible reason 2: simply due to the difference of sample sizes

Furthermore, there may also be another possibility where the reduced data indeed reflects the same information as the full data. In other words, the information carried by the reduced data "looks" different from that carried by the full data only by chance. In such case, the difference between the two results (in terms of the significance of `exper`) may be simply due to the different sample sizes, that is, the smaller sample size of the reduced data causes greater variance of the coefficient estimators and hence the coefficient for `exper` is shown not statistically significant for the reduced data.

2 Problem 2

2.1 What would you say to him?

Such question should be answered under two cases.

Case 1: the model assumed is correct

I would say that the effect of each independent variable are indeed statistically significant under significance level $\alpha = 0.1$. In other words, one can say that there is about 90% of confidence to believe that the confidence interval for each of the parameters does not contain 0 and hence the effects of the independent variables are significant under the precision of data itself. The very low R^2 value 0.092 only tells that the pure variation of the dependent variable is too big in nature so that only very small proportion of it can be explained by those "significant" independent variables.

Case 2: the model assumed is not correct

I would say that the effect of each independent variable may not be truely statistically significant under significance level $\alpha = 0.1$ if there actually exist some useful independent variables outside the model. What make it worse is when such useful variables are highly correlated with those included in the model. Under such situation, some of the effects of the variables outside the model may be somehow absorbed and thus mis-attributed to those independent variables included in the model. Moreover, the R^2 value 0.092 is basically meaningless in terms of the goodness-of-fit since the model assumed is incorrect, and hence nothing is worth commenting in regards with the relation between it and the significance.

Remark

However, it is usually quite difficult to decide whether or not to believe that the model assumed is (almost) correct. It is believed that domain knowledge is necessary to make such decision.

2.2 What are the possible reasons that can cause the significance?

The possible reasons that can cause the significance are either that the sample size is very large which brings up high precision of the data, or that the effect sizes of those independent variables are indeed physically large. Moreover, we should also not exclude the possibilities under which the data was suffered too much by the random disturbances and hence type I error occurred. If this is the case, it is sorry to say that the researchers were too unlucky.